Team Details & Problem Statement

PROBLEM STATEMENT

Dark Web Watchdog

AI-Powered Dark Web Threat Intelligence & Leak Detection Platform

A proactive security solution designed to monitor compromised data marketplaces, identify corporate and personal leaks in real-time, and deliver automated mitigation recommendations.

TEAM DETAILS

TEAM NAME

HACKABEES

TEAM LEADER

PRANAY T G

TEAM MEMBERS

AATHIF M C

CLIVE CLEMENT E

INSTITUTION

JAIN DEEMED TO BE UNIVERSITY

BENGALURU, INDIA

Idea / Approach Details

THE PROBLEM

Every day, millions of critical assets are leaked & sold on dark web marketplaces.

- User credentials & API keys
- Credit card & personal details
- Proprietary company documents

CURRENT CHALLENGES

- **Lack of Awareness:** Users don't know when their data has been leaked.
- **High Barriers:** Small businesses cannot afford enterprise threat intelligence.
- **Delayed Action:** Data breaches are detected far too late.
- **Fragmented Tools:** No unified platform for monitoring, alerts & AI recommendations.

FLOW CHART :



Technology Stack & Use Cases

TECHNOLOGY STACK

FRONTEND

React • Tailwind CSS • TypeScript • Framer Motion

BACKEND & DATABASE

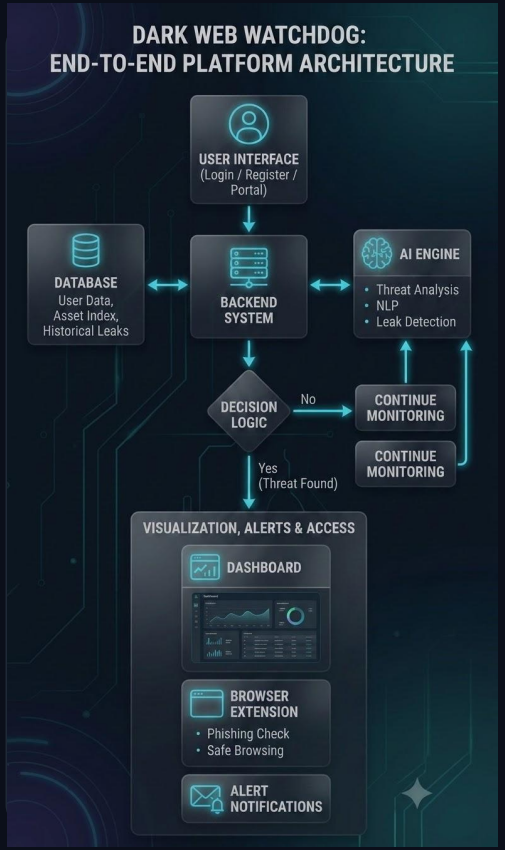
FastAPI • Python • PostgreSQL

AI / NLP ENGINE

OpenAI / Llama • spaCy • Sentence Transformers

EXTENSION & ALERTS

Chrome API • JavaScript • MV3 • Email • Push



TARGET USE CASES

Individual Users

Monitor credentials & personal identity leaks.

Students

Academic threat analysis and cyber research.

Small Businesses

Affordable, vital brand and asset monitoring.

Enterprises

Robust threat intelligence & scalable web extensions.

Effort, Cost & Timeline

WEEK 1 • INCEPTION

- UI Design
- Backend Setup
- Database

WEEK 2 • CORE DEV

- AI Integration
- Leak Detection
- Dashboard

WEEK 3 • EXTENSION

- Browser Ext.
- Alerts System
- Reports Panel

WEEK 4 • RELEASE

- Testing & QA
- Deployment
- Final Demo

Estimated Cost

Item	Cost
Cloud Hosting	$0–20/month (prototype)
Database	Free Tier
AI API	Free/Credits
Email Service	Free Tier

Team Effort

Role	Responsibility
Frontend	Dashboard & UI
Backend	APIs & Database
AI	Threat Detection
Extension	Browser Protection

Dependencies, Challenges & Future Scope

DEPENDENCIES

- Public breach datasets
- AI models
- Email notification service
- Chrome Extension APIs

CHALLENGES

- Ethical & legal constraints on dark web monitoring
- False positives in phishing detection
- Real-time data availability
- API rate limits

FUTURE SCOPE

- Live Threat Intelligence
- Ransomware Detection
- Mobile Application
- SOC & SIEM Dashboard
- Enterprise Monitoring
- AI Threat Prediction
- Global Threat Heatmaps
- Zero-Day Threat Detection

Business Potential

SaaS Platform

Subscription plans:

- Free
- Premium Individual
- Small Business
- Enterprise

Customers

- Individuals
- Colleges
- SMEs
- Banks
- IT Companies
- Government Agencies

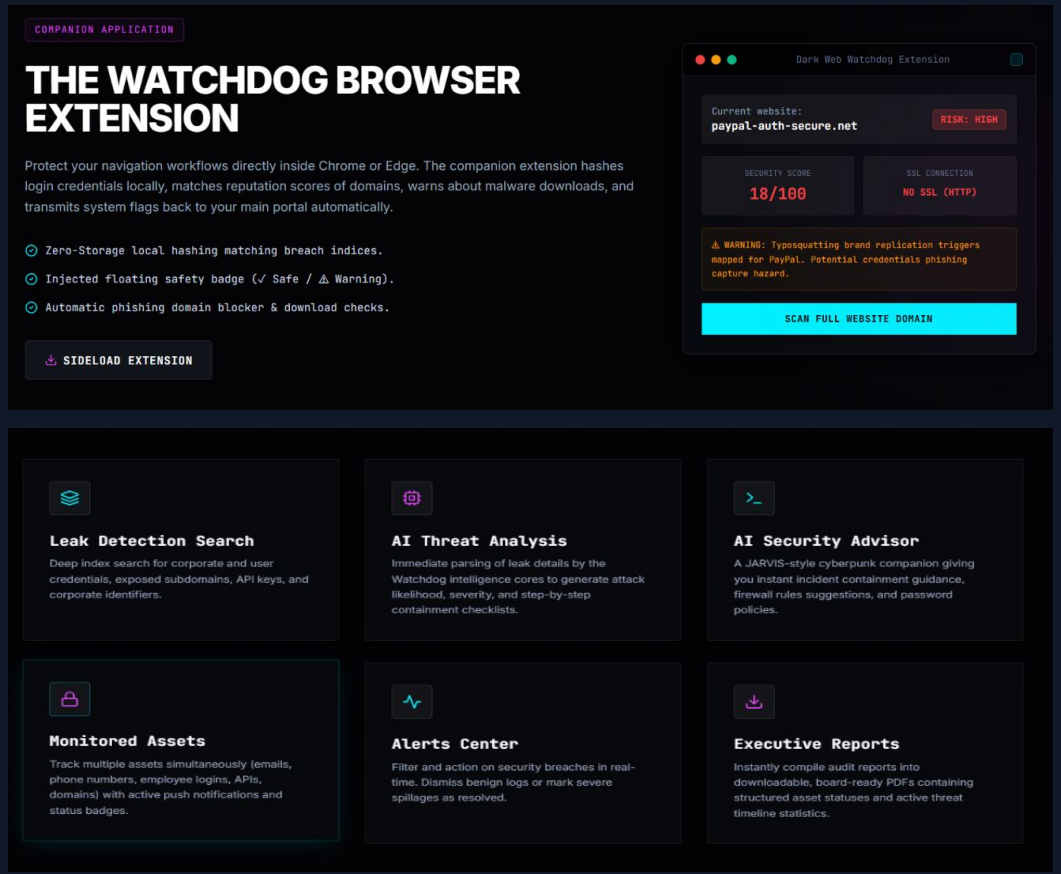
Revenue

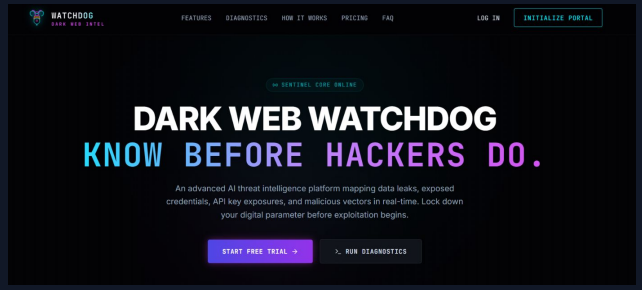
- Monthly subscriptions
- Enterprise licensing
- API access
- Security consulting

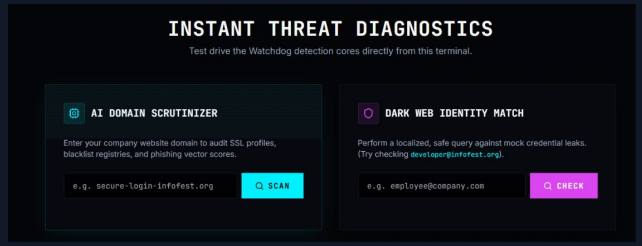
UI Mockups

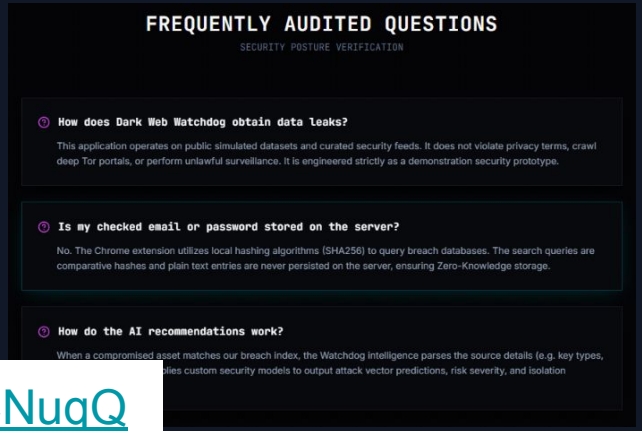
- Dashboard
- Threat Timeline
- AI Security Advisor
- Alerts Page
- Browser Extension Popup

Application Prototype









Prototype demonstration link : https://youtu.be/9FQgPqCNuqQ

Conclusion

Dark Web Watchdog – Securing Digital Identities with AI

KEY TAKEAWAYS

- AI-powered platform for proactive cyber threat detection.
- Detects exposed credentials and sensitive information using simulated/public breach intelligence.
- Provides real-time alerts, risk scoring, and actionable security recommendations.
- Integrated browser extension enhances protection against phishing and malicious websites.
- Designed to be scalable for individuals, businesses, educational institutions, and enterprises.

IMPACT

"Dark Web Watchdog empowers users to identify cyber risks early, respond quickly, and strengthen their digital security through intelligent monitoring and AI-driven insights."

VISION

"Protecting Digital Identities Before It's Too Late."

Thank You

QUESTIONS & FEEDBACK